

Report R27 — the evolution operator’s Schmidt rank

What the literature establishes, and why a linear-dimensional module forces linear growth

rev. 2: the bound is now two-sided, and the law is exact

Tier 1 analysis

2026-08-23

Abstract

Rule 156 has an operator Schmidt rank $\chi(t)$ of its *evolution operator* that grows linearly in t , in a clean, disorder-free, translation-invariant Floquet circuit. This report establishes what the literature says about that observable and supplies a mechanism. §1 is an audited pass over the twelve references supplied plus eleven found by search, sorted by relevance, each with the finding that bears on our claim. **The baseline is Dubail’s:** the OSEE of $U(t) = e^{-iHt}$ grows *linearly*, hence χ *exponentially*, “unless the Hamiltonian is in a localized phase”. Zhou and Luitz classify the three observed growth laws for $U(t)$ — linear, power law, logarithmic — and both sub-linear classes occur only with disorder. Every disorder-free sub-linear result we could find, including Alba–Dubail–Medenjak for Rule 54 and the very recent Tan–Prosen clean non-integrable circuit, is about *local operators in the Heisenberg picture*, which Dubail explicitly separates from the propagator. We found no disorder-free system with sub-linear evolution-operator OSEE. §2 then derives the mechanism. The key quantity is not integrability but the **number of defects that can cross the cut**: if the sector containing the cut is spanned by configurations of p domain walls with p bounded, then $\chi(t) = O(t^p)$, and if p is extensive χ grows exponentially. The kink module of W_{156}/W_{198} has $p = 1$, giving $\chi(t) = O(t)$; the hard-core module of W_{201}/W_{108} has extensive p , giving exponential growth. Fragmentation into modules of polynomial dimension is therefore a disorder-free mechanism for the localization signature.

New in rev. 2. §3 closes the gap left open in rev. 1. The upper bound is sharpened to an exact combinatorial statement — a *last-passage identity* valid for any operator and any splitting, giving $\text{rank}(P_R A^t P_L) \leq t \cdot \text{rank}(P_R A P_L)$ — and the cross transfer of the kink walk is shown to be *rank one*, in a cell basis in which the walk is a two-band chiral quantum walk. A matching lower bound follows from an explicit rank- t factorisation whose two vector families are independent by causality. The result is the two-sided bound $2t \leq \chi_{\text{module}}(t) \leq 2t + 2$, so $\chi = \Theta(t)$, with the measured value exactly $2t + 1$ for both rules, every chain length and every cut tested. For the *full* chain the exact law is $\chi(t) = 6t - 7$ for $4 \leq t \leq N/2$, verified at $N = 8, 10, 12, 14, 16$; the slope 6 itself is recorded as measured, together with three ruled-out derivations of it.

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1 Protocol: the literature, sorted by relevance

Two objects must be kept apart throughout, because the literature treats them very differently and conflating them is the trap this project already fell into once:

- **(L)** the operator entanglement of a *local* operator $O(t) = U^\dagger O U$ in the Heisenberg picture;
- **(U)** the operator entanglement — equivalently the Schmidt rank of the Choi vector, equivalently the MPO bond dimension — of the *evolution operator* $U(t)$ itself.

Our measurement is **(U)**. Sub-linear results for **(L)** do not bear on it.

1.1 Tier 1: directly on the claim

reference	object	finding that bears on our claim
Dubail, J. Phys. A 50 , 234001 (2017)	(U)	The decisive baseline. “The OSEE of the evolution operator $U(t) = e^{-iHt}$ increases linearly with t , unless the Hamiltonian is in a localized phase.” Explicitly contrasts this with local operators, whose OSEE “grows sublinearly (perhaps logarithmically)”. Linear OSEE $\Rightarrow \chi$ exponential. <i>Localization is named as the only exception.</i>
Zhou & Luitz, PRB 95 , 094206 (2017)	(U)	Classifies the growth of operator EE <i>of the time evolution operator</i> across chaotic, MBL and Floquet systems: linear, power-law and logarithmic growth are all observed, before extensive saturation. Crucially the two sub-linear classes are the <i>random-field</i> Heisenberg model — power law at weak disorder, logarithmic at strong disorder. The most chaotic Floquet model saturates at the Page value.
Thakur, Pain & Roy, PRB 111 , 174206 (2025)	(U)	Operator entanglement of the time-evolution operator in the MBL phase, showing a <i>logarithmic light cone</i> , derived from eigenstate correlations without invoking ℓ -bits. This is the closest published phenomenology to ours — same object, same functional form — but obtained <i>from disorder</i> .

reference	object	finding that bears on our claim
Prosen & Žnidarič, PRE 75 , 015202 (2007)	(L)/state	The origin of the exponential-versus-polynomial dichotomy for MPO bond dimension: $D_\epsilon(t) \propto e^{ct}$ in quantum chaotic systems, polynomial when integrable. <i>But the evolved objects are observables and thermal states, not the propagator</i> , so this does not license “integrable $\Rightarrow$ polynomial χ ” for $U(t)$.

1.2 Tier 2: the disorder-free sub-linear results, all for local operators

reference	object	finding
Alba, Dubail & Medenjak, PRL 122 , 250603 (2019)	(L)	Logarithmic upper bound on operator entanglement for <i>all local operators</i> in the Rule 54 chain, and evidence that log growth is generic for interacting integrable systems. A cellular automaton, so tempting — but the wrong object.
Tan & Prosen, arXiv:2603.19363 (2026)	(L)	“Logarithmic growth of operator entanglement in a <i>clean non-integrable</i> circuit”. Semi-ergodic dual-unitary brickwork; mechanism is that the Heisenberg evolution of a single traceless single-site operator lies in a restricted subspace. The nearest disorder-free precedent in spirit — log growth from a restricted subspace rather than from integrability or disorder — but again for a local operator, not $U(t)$.
Prosen & Pizorn, PRA 76 , 032316 (2007)	(L)	Transverse Ising chain: OSEE of local operators <i>saturates</i> for finite Majorana index and grows <i>logarithmically</i> for infinite index. Free fermions; local operators.
Pal & Lakshminarayan, PRB 98 , 174304 (2018)	(U)	Entangling power of time-evolution operators, integrable versus non-integrable. The integrable kicked transverse-field Ising chain still shows <i>ballistic</i> growth of operator von Neumann entropy; its entangling power saturates lower than the non-integrable case but the growth is not sub-linear. <i>Integrability alone does not buy sub-linear $\chi(t)$.</i>

1.3 Tier 3: framework, and the closest methodological neighbour

reference	object	finding
Merbis & Bakker, ICCS 2024, arXiv:2406.04895	(U), classical	The closest method: builds the MPO of a cellular automaton’s transition matrix and classifies elementary CA by operator-entanglement growth. Types I (converges to a constant, e.g. their rules 136, 108), II (rises then falls, peak at $t \sim L$), III (linear growth to an L -dependent plateau, chaotic-like). Generic bond dimension quoted as 4^t , i.e. $S(t) = 2t$. <i>No logarithmic class appears</i> . Their object is the classical transition matrix of a deterministic CA, not a quantum evolution operator with Hadamards, so the rule numbers are not comparable to ours despite the coincidence.
Zanardi, Zalka & Faoro, PRA 62 , 030301 (2000); Zanardi, PRA 63 , 040304 (2001)	(U)	Where the object comes from: entanglement of a unitary regarded as a vector in Hilbert–Schmidt space, and the entangling power. Definitional; sets the measure we and everyone else use.
Jonay, Huse & Nahum, arXiv:1803.00089 (2018)	(L)/(U)	The membrane / line-tension coarse-graining. Predicts <i>linear</i> growth for generic chaotic dynamics with rate set by the surface tension $\mathcal{E}(v)$. Our result sits outside this picture, which is the point.
Hahn, Luitz & Chalker, PRX 14 , 031029 (2024)	(U)	Eigenstate correlations of the time-evolution operator beyond ETH; uses Floquet models without conserved densities to isolate scrambling. Provides the eigenstate-correlation language in which Thakur–Pain–Roy then derive the MBL log light cone.
Bandyopadhyay & Lakshminarayan, arXiv:quant-ph/0504052 (2005)	(U)	Entangling power of quantum chaotic evolutions; early evidence that chaotic evolutions approach the random-matrix value.
Musz, Kuś & Życzkowski, PRA 87 , 022111 (2013)	(U)	Geometry of unitary gates, perfect entanglers, unistochastic maps. Fixes what “maximal” means for the measure.
arXiv:2603.05656 (2026), <i>Classical simulability from operator entanglement scaling</i>	(L)	Simulability classification by the scaling exponent α : $\alpha \geq 1$ blocks efficient MPO approximation, $\alpha < 1$ permits it over restricted state ensembles. Local-operator entanglement.

1.4 Verdict of the pass

1. For the *evolution operator*, the established law is linear OSEE, i.e. **exponential** $\chi(t)$, with exactly one documented exception: **localization**, which requires disorder (Dubail; Zhou–Luitz; Thakur–Pain–Roy).
2. Integrability is *not* an exception for this object. Prosen–Žnidarič’s polynomial result is for observables and states; Pal–Lakshminarayan find ballistic operator-entropy growth even for an integrable kicked Ising chain.
3. Every disorder-free sub-linear result we found — Alba–Dubail–Medenjak, Prosen–Pižorn, Tan–Prosen — is for *local operators*, which Dubail explicitly separates from the propagator.

4. We found **no** report of sub-linear evolution-operator Schmidt-rank growth in a clean, disorder-free system. A linear $\chi(t)$ for $U(t)$ in a translation-invariant Floquet circuit is, on this evidence, unoccupied ground.

Remark 1. The nearest conceptual precedent is Tan–Prosen: log growth in a clean non-integrable circuit because the evolved operator is confined to a *restricted subspace*. That is the same shape of mechanism as ours — confinement to a small module rather than integrability or disorder — applied to the other object. It should be cited as the closest relative and distinguished on the object.

2 The mechanism: defect counting bounds the Schmidt rank

We now give the argument that a module of linear dimension forces linear $\chi(t)$. The statement turns out not to be about dimension as such, but about *how many defects can cross the cut*.

2.1 Setup

Let U be one period of a brick-wall circuit with strict light-cone velocity v (sites per period). Fix a cut at bond x and let $\chi_x(t)$ be the operator Schmidt rank of $U(t) = U^t$ across it — equivalently the rank of the Choi vector $|U(t)\rangle\rangle$ under the bipartition, equivalently the MPO bond dimension at that bond.

The generic bound is the light cone: $U(t)$ acts as the identity outside a causal diamond of width $2vt$ about the cut, so $\chi_x(t) \leq 4^{2vt}$. Everything in §1 that grows exponentially is saturating a bound of this form. To do better one needs the dynamics to explore only a small part of the operator space on the diamond.

2.2 The kink chain

Take W_{156} . Its sectors are wall-free gaps, and inside a gap the state space is spanned by the *kink states*

$$|k\rangle = \underbrace{1 \cdots 1}_k \underbrace{0 \cdots 0}_{l-k}, \quad k = 0, \dots, l,$$

of dimension $l+1$ — one degree of freedom, the position of a single domain wall. Restricted to the gap,

$$U(t)|_{\text{gap}} = \sum_{k,k'} A_{kk'}(t) |k\rangle\langle k'|,$$

a *single-particle propagator*. Two facts make the Schmidt rank collapse.

Every kink state is a product across the cut. Cutting at x ,

$$|k\rangle = |L_k\rangle \otimes |R_k\rangle, \quad \begin{cases} L_k = 1^k 0^{x-k}, & R_k = 0^{l-x} & k \leq x, \\ L_k = 1^x, & R_k = 1^{k-x} 0^{l-k} & k > x. \end{cases}$$

So each term $|k\rangle\langle k'|$ is already a product operator $\ell_{kk'} \otimes r_{kk'}$ with $\ell_{kk'} = |L_k\rangle\langle L_{k'}|$ and $r_{kk'} = |R_k\rangle\langle R_{k'}|$. No entanglement is generated by the basis itself.

Three of the four index sectors are rank one. Split the sum by the side of the cut:

- $k, k' \leq x$: then $R_k = R_{k'} = 0^{l-x}$, so *every* term shares the same right factor. All of these together contribute Schmidt rank ≤ 1 .
- $k, k' > x$: then $L_k = L_{k'} = 1^x$, the same left factor. Rank ≤ 1 .

- $k \leq x < k'$ and $k' \leq x < k$: the crossing sectors. Here the left factor is $|L_k\rangle\langle 1^x|$, labelled by k alone, and the right factor is $|0^{l-x}\rangle\langle R_{k'}|$, labelled by k' alone. The contribution is the rank of the coefficient block $A_{kk'}$ restricted to that range.

Causality restricts the crossing blocks: $A_{kk'}(t) = 0$ unless $|k - k'| \leq 2vt$, and a term contributes across the cut only if the kink is within the light cone of x , i.e. $|k - x|, |k' - x| \lesssim vt$. Each crossing block is therefore at most $(vt) \times (vt)$, of rank at most vt . Hence

$$\boxed{\chi_x(t) \leq 2 + 2vt = O(t)}. \quad (1)$$

2.3 The general statement

Nothing above used δ , the Hadamard, or integrability. What was used is that the sector is labelled by the position of *one* defect.

Proposition 1 (Defect counting). *Suppose U preserves a sector whose basis states are labelled by the positions of p defects on the chain, each defect state being a product state across any cut, and suppose U has strict light-cone velocity v . Then the operator Schmidt rank of $U(t)$ across a cut obeys*

$$\chi_x(t) = O((vt)^p).$$

If instead the number of defects is extensive in the region size, the bound degrades to the light-cone bound and $\chi_x(t)$ may grow exponentially.

Sketch. As above: a configuration is fixed by the p defect positions, so a basis operator $|c\rangle\langle c'|$ factorises across the cut with left factor determined by the defect positions to the left and right factor by those to the right. Terms in which no defect lies within the light cone of the cut share a common factor on one side and contribute rank 1. Terms with $j \leq p$ defects inside the light cone are labelled by at most $(2vt)^j$ positions on each side, so the corresponding coefficient block has rank at most $(2vt)^j$. Summing over $j \leq p$ gives $O((vt)^p)$. $\square$

2.4 The two families

rules	gap module	dimension	defects p	predicted $\chi(t)$
W_{156}, W_{198}	kink chain $1^a 0^b$	$l + 1$, linear	1	$O(t)$, linear
W_{201}, W_{108}	hard-core chain	$\sim \varphi^l$, exponential	extensive	exponential

This reproduces both measured behaviours: linear $\chi(t)$ for rule 156, exponential for 201/108. It also explains why the split is so sharp despite the two families differing by a single letter of the gate word: the V -context decides the wall word, the wall word decides whether a gap admits an ascent, and that decides whether the gap carries one defect or extensively many.

Remark 2 (What the argument does and does not give). It gives linearity, and the identification of the exponent with the defect number. It does *not* give the measured slope 6. It is an upper bound, so it does not by itself prove the growth is not slower — §3 supplies the matching lower bound. And it assumes the sector basis is product across the cut, which holds for kink and hard-core configurations but must be checked for any other module.

Remark 3 (Why this is the localization signature without localization). In an MBL system the log light cone comes from an emergent set of quasi-local integrals of motion which confine information transport. Here the confinement is kinematic: the sector simply has too few configurations within the light cone for the Schmidt rank to grow. Fragmentation supplies, without disorder, the thing localization supplies with it — a bound on how many distinguishable histories can cross a cut in time t . That is the sense in which Proposition 1 is the mechanism the literature pass says is missing.

3 The tight bound

Proposition 1 gives $\chi = O(t)$. This section replaces it, for the kink module, by a two-sided bound with an exactly matching measurement. All numbers below come from `analytics/kink_schmidt.py`, which rebuilds the propagator from the project's exact $\mathbb{Z}[1/\sqrt{2}]$ branch machinery, not from any earlier tabulation. One bookkeeping point: at obc0 the maximal one-wall Krylov sector of an N -site chain is $\{|k\rangle = 1^k 0^{N-k} : k = 1, \dots, N\}$, of dimension $l = N$ — the all-zero configuration is frozen and sits outside it — so l and N coincide throughout this section. Closure is exact: the amplitude leaking out of the module is 0.0, not merely small.

3.1 Step 1: the reshuffle collapses onto the coefficient matrix

Fix the cut at bond x and recall from §2 that $|k\rangle = |L_a\rangle \otimes |R_b\rangle$ with $a = \min(k, x)$, $b = \max(k, x)$. The pair (a, b) determines k , so the operator Schmidt decomposition of $U(t)|_{\text{module}}$ is nothing but a re-indexing of the coefficient matrix A^t : rows (a, a') , columns (b, b') . The four index sectors land as follows.

index sector	row	column	content
$k, k' \leq x$	(k, k')	(x, x)	the whole LL block, in one column
$k, k' > x$	(x, x)	(k, k')	the whole RR block, in one row
$k \leq x < k'$	(k, x)	(x, k')	the crossing block C_{LR}
$k' \leq x < k$	(x, k')	(k, x)	the crossing block C_{RL}

The two crossing blocks occupy disjoint rows and disjoint columns, and the remaining content is confined to the single row (x, x) and the single column (x, x) . Hence, with no approximation,

$$\text{rank } C_{LR} + \text{rank } C_{RL} \leq \chi_x(t) \leq 2 + \text{rank } C_{LR} + \text{rank } C_{RL}. \quad (2)$$

The whole problem is now the rank of one crossing block.

3.2 Step 2: a last-passage identity, and the transfer rank

Proposition 2 (Transfer rank). *Let A act on a finite-dimensional space and let $P_L + P_R = \mathbb{I}$ be complementary projectors. Then for every $t \geq 1$*

$$P_R A^t P_L = \sum_{s=0}^{t-1} (P_R A P_R)^{t-1-s} (P_R A P_L) (P_L A^s P_L), \quad (3)$$

and consequently

$$\text{rank}(P_R A^t P_L) \leq t \cdot \text{rank}(P_R A P_L).$$

Proof. Expand A^t as a sum over sequences of L/R labels, one per time step. Every sequence contributing to $P_R A^t P_L$ starts in L and ends in R , so it has a unique *last* time s at which the label is L , with $0 \leq s \leq t-1$. Grouping the sequences by that s gives exactly (3): the factor $P_L A^s P_L$ collects all histories up to time s (unrestricted in between), the factor $P_R A P_L$ is the step out of L at time $s+1$, and $(P_R A P_R)^{t-1-s}$ the remaining steps, which stay in R by construction. Each summand contains $P_R A P_L$ as a factor, so each has rank at most $\text{rank}(P_R A P_L)$, and rank is subadditive. $\square$

The identity is checked numerically to $\leq 2.2 \times 10^{-16}$ for the kink propagator at $N = 24, 30$ and $t \leq 10$, and for a random 10×10 unitary; it is elementary, but it is the step that converts a *static*, one-period quantity into a linear-in- t bound.

It remains to compute $\text{rank}(P_R A P_L)$ for the kink walk. Group the kink positions into cells $\{2j-1, 2j\}$ and pass to

$$|p_j\rangle = \frac{1}{\sqrt{2}}(|2j-1\rangle + |2j\rangle), \quad |m_j\rangle = \frac{1}{\sqrt{2}}(|2j-1\rangle - |2j\rangle).$$

In this basis the exact one-period propagator of W_{156} reads, in the bulk,

$$\begin{aligned} A|p_j\rangle &= \frac{1}{2}(|m_{j-1}\rangle + |m_j\rangle - |p_j\rangle + |p_{j+1}\rangle), \\ A|m_j\rangle &= \frac{1}{2}(|m_{j-1}\rangle - |m_j\rangle - |p_j\rangle - |p_{j+1}\rangle), \end{aligned} \quad (4)$$

i.e. a two-band quantum walk hopping by at most one cell per period. Its inter-cell transfer operators are

$$T_+ = \frac{1}{2}|p\rangle(\langle p| - \langle m|), \quad T_- = \frac{1}{2}|m\rangle(\langle p| + \langle m|), \quad (5)$$

both of rank one (W_{198} is the mirror of W_{156} and reflection sends $k \mapsto l+1-k$, so its cell tiling is the shifted pairing $\{2j, 2j+1\}$ and the roles of p and m are exchanged). The coin state p is the sole right-moving channel and m the sole left-moving one: the walk is chiral, with one channel per chirality per cell. Measured directly, $\text{rank}(P_R A P_L) = 1$ at *every* cut $x = 1, \dots, l-1$, for both rules at $N = 12, 20, 24$, so Proposition 2 gives $\text{rank } C_{LR} \leq t$, likewise for C_{RL} .

3.3 Step 3: the matching lower bound

Because $P_R A P_L = ab^\top$ is rank one, (3) is an explicit rank- t factorisation:

$$P_R A^t P_L = \sum_{s=1}^t u_s v_s^\top, \quad u_s = (P_R A P_R)^{t-s} a, \quad v_s = (P_L A^{s-1} P_L)^\top b.$$

If $\{u_s\}_{s \leq t}$ and $\{v_s\}_{s \leq t}$ are each linearly independent, the factorisation is $UV^\top$ with both factors of full column rank t , so the rank is exactly t . Independence follows from the light cone: by (5) the rightward amplitude per period is exactly $\frac{1}{2}$ and never vanishes, so u_s reaches exactly $t-s$ cells to the right of the cut with leading amplitude $2^{-(t-s)}$, and v_s exactly $s-1$ cells to the left. The reaches are pairwise distinct, hence each family is independent. Measured: $\dim \text{span}\{u_s\} = \dim \text{span}\{v_s\} = \text{rank} = t$ for all $t \leq 9$ at $N = 24, 30$, both rules, with the factorisation reproducing $P_R A^t P_L$ to $\leq 2.2 \times 10^{-16}$.

3.4 The law

Combining (2) with $\text{rank } C_{LR} = \text{rank } C_{RL} = t$:

Theorem 1 (Kink module). *For W_{156} and W_{198} on their maximal one-wall module of dimension l , and any cut x with $1 \leq t < \min(x, l-x)$ — i.e. as long as the light cone has not reached either end of the chain —*

$$2t \leq \chi_x(t) \leq 2t+2, \quad \text{so} \quad \chi_x(t) = \Theta(t).$$

The measured value is the middle one, and the saturation is a clean truncation:

$$\chi_x(t) = \min(2t+1, \chi_x^{\text{sat}}), \quad t \geq 1, \quad (6)$$

with χ_x^{sat} a t -independent, cut-dependent ceiling. Verified exhaustively for both rules at $N = 12, 16, 20$, *every* cut $x = 1, \dots, N-1$ and t up to $3N$, and at $N = 24, 30$ for $x = 3, N/2, N/2+1$. At $t = 0$ the value is 2: the identity on the module is not a product operator across the cut, being $\mathbb{K}_{\text{left}} \otimes P_R + P_L \otimes \mathbb{K}_{\text{right}}$. This is the statement rev. 1 left open. The ceiling χ_x^{sat} is a finite-chain edge effect, small when the cut is near an end; no closed form for it is claimed here.

3.5 The full chain

On the full 2^N space the exact operator Schmidt rank of U^t across the middle bond is

$$\boxed{\chi(t) = 6t - 7, \quad 4 \leq t \leq N/2.} \quad (7)$$

Measured series (identical for W_{156} and W_{198} , obc0, $x = N/2$):

N	$\chi(t), t = 0, 1, 2, \dots$	law holds for	first deviation
8	1, 4, 8, 12, 17, 20, 21, ...	$t \leq 4$	$t = 5$
10	1, 4, 8, 12, 17, 23, 27, 30, 31, ...	$t \leq 5$	$t = 6$
12	1, 4, 8, 12, 17, 23, 29, 33, 37, 41, ...	$t \leq 6$	$t = 7$
14	1, 4, 8, 12, 17, 23, 29, 35, 39, 43	$t \leq 7$	$t = 8$
16	1, 4, 8, 12, 17, 23, 29, 35, 41, 45, 49	$t \leq 8$	$t = 9$

The increments are 3, 4, 4, 5 and then a constant 6; the deviations past $t = N/2$ are the finite-chain saturation, and the window grows by one period per two sites, exactly as a light cone should. The $N = 14$ and $N = 16$ rows were computed with a sparse propagator and a randomised rank estimator (oversampled to 160 probe vectors, so the reported rank is exact whenever it is below the cap, which it always is).

Remark 4 (The slope 6 is not yet derived — and three routes are closed). Theorem 1 explains the linearity and gives slope 2 per straddling gap. The factor 3 relating that to (7) is not derived here, and the obvious derivations do not work:

1. *Not a sum over Krylov sectors.* The reshuffle is supported on entry (y, y') only when y, y' share a sector, so it splits into blocks of disjoint support; but the left Schmidt vectors are shared *between* sectors, and the sum is wildly larger than the total. At $N = 12$ (377 sectors) $\sum_S \chi_S(t) = 473, 569, 700, 747, 763, 749, 731, \dots$ against $\chi(t) = 1, 4, 8, 12, 17, 23, 29, \dots$ — not even monotone. Fragmentation alone does not make the MPO small; the sharing does.
2. *Not a sum over growing sectors either.* Only 12 of the 377 sectors have a χ_S still growing at $t = 6$, each with slope 2 while unsaturated, for a total of 24 where the full chain increments by 6.
3. *Not an accumulating channel basis.* The left Schmidt subspace does not grow as a filtration: projecting the $t+1$ subspace against the t one leaves 20–21 new directions out of 23–29, so the subspace rotates almost completely each period and there is no fixed set of “six new channels” to count. Every Schmidt vector also acts non-trivially out to the far end of the half-chain, so the new directions cannot be localised at the light-cone edge.

The slope is therefore a genuinely global count over the wall grammar, and deriving it needs the MPO of U^t itself rather than any sector-wise argument.

4 What would settle it

1. **The slope 6.** The one quantitative gap left. (7) is exact over the whole accessible window and the three cheap explanations above are excluded, so this needs the transfer-matrix structure of the MPO of U^t .
2. $\chi(t)$ **for** W_{201}, W_{108} . Held by the experimental side of the project; the linear-versus-exponential contrast is the whole claim and Proposition 1 predicts exponential growth there because the hard-core module has extensive defect number.

3. **Beyond one straddling gap.** Theorem 1 is the $p = 1$ case. Confirming the $O(t^p)$ prediction of Proposition 1 on a module with $p = 2$ would turn defect counting from an explanation of two data points into a law.

References

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